Lecture 16: Decision Tree

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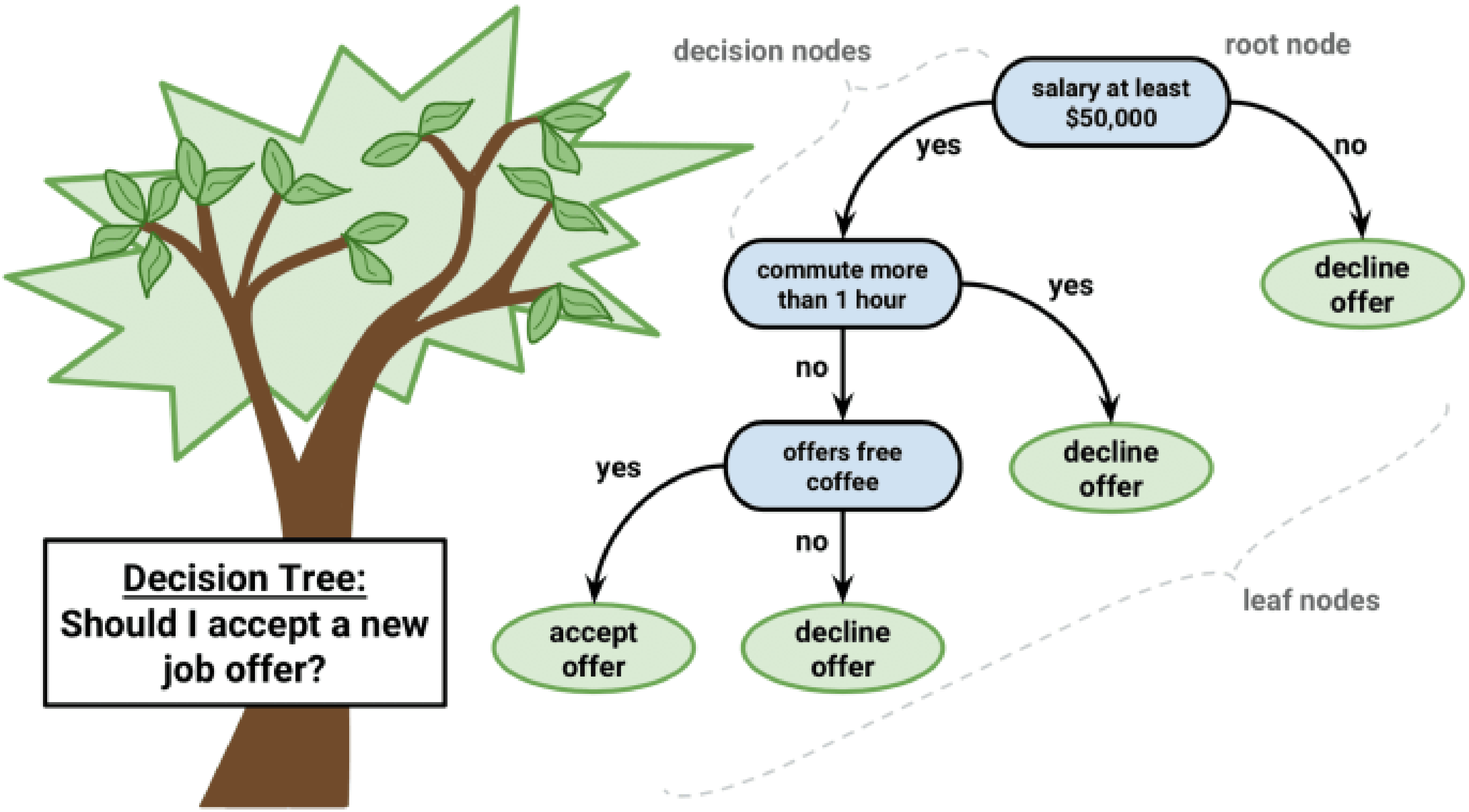
Decision Trees: Example



Decision Trees

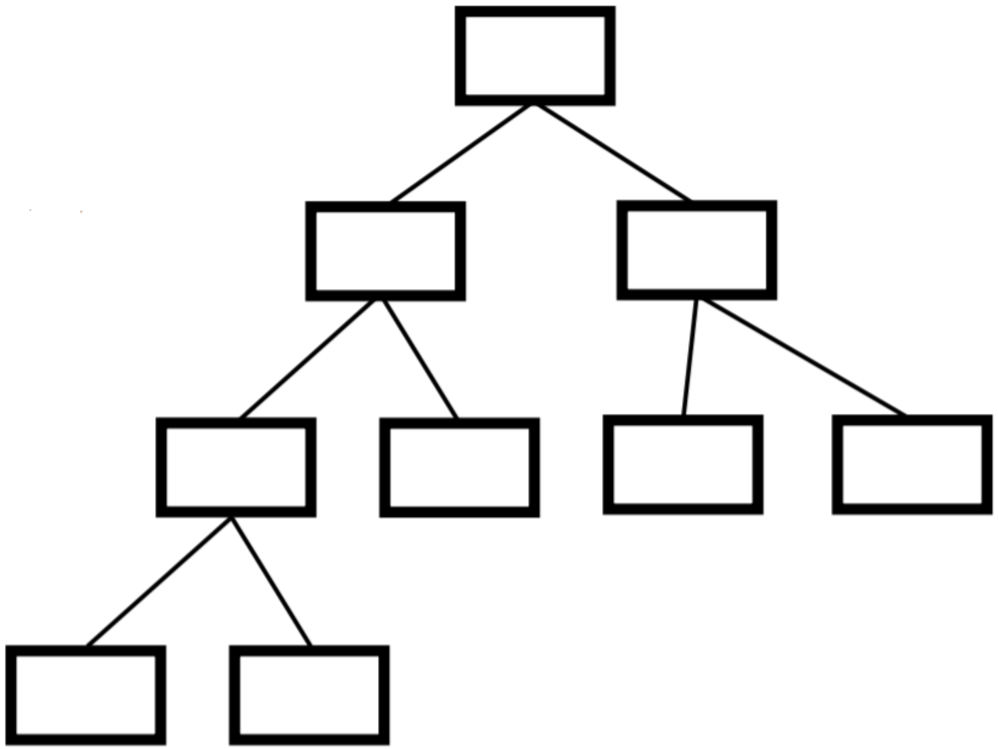
- *Decision Trees* are versatile Machine Learning algorithms that can perform both classification and regression tasks, and even multioutput tasks
- Decision Trees are also the fundamental components of Random Forests which are among the most powerful Machine Learning algorithms available today
- One of the many qualities of Decision Trees is that they require very little data preparation. In fact, they don't require feature scaling or centering at all

Decision Trees

- A Decision Tree is a flow-chart-like tree structure
  - **Internal nodes**: test an attribute
  - **Branches:** test result
  - **Leaf nodes**: Class label

- At each node, one feature is picked to split training examples into distinct classes

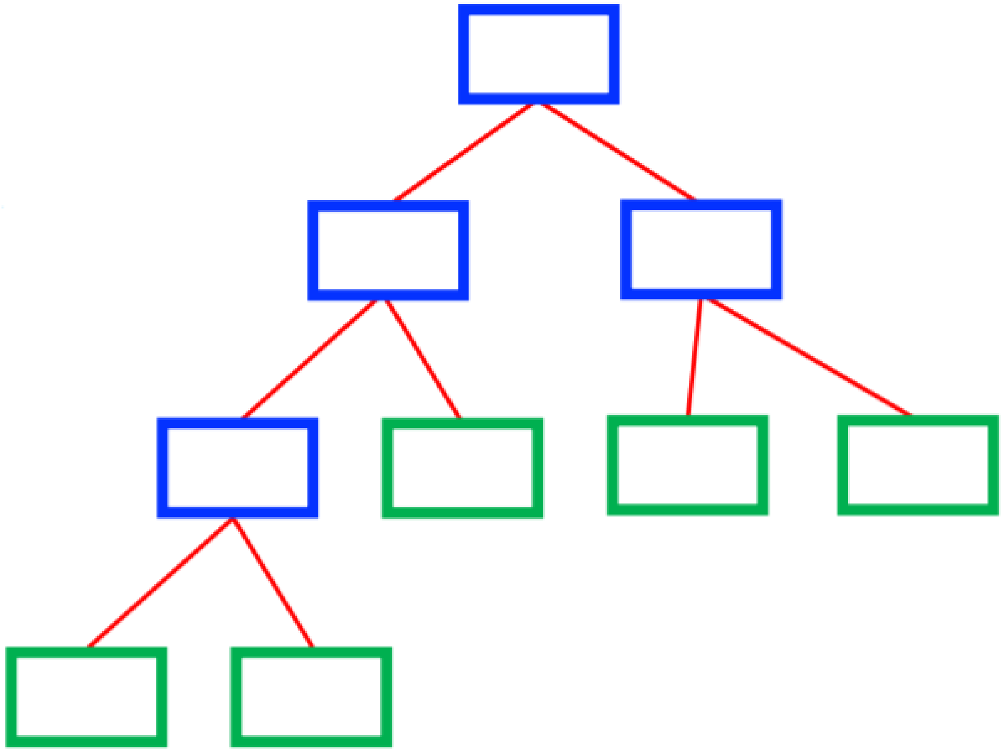
- New samples are classified by following a matching path to a leaf node



Decision Tree Methods

- Goal: Predict Class label
  - Internal nodes: test an attribute
  - Branches: test result
  - Leaf nodes: Class label

- How to build(i.e. train) Decision Trees? Two major methods are CART or ID3
  - CART → Classification and Regression Trees, Gini index used for best attribute
  - ID3 →Iterative Dichotomiser 3, Entropy used for best attribute



How to determine the Best Split

- Purpose of splitting: Make homogeneous/pure groups

- Greedy approach: approach for solving a problem by selecting the best option available at the moment
  - Nodes with purer class distribution are preferred
  - Need a measure/metric of node impurity, e.g for CART, Gini impurity measure is being used

Gini impurity

- Gini impurity:

$$G_i = 1 - \sum_{k=1}^{n} {P_{i,k}}^2$$

- Here, $P_{i,k}$ *is the ratio of class k instances among the training instances in the* i^{th} *node*

- a node's gini attribute measures its *impurity: a node is* "pure" (gini=0) if all training instances it applies to belong to the same class

The CART Training Algorithm

- Scikit-Learn uses the *Classification and Regression Tree (CART) algorithm to train* Decision Trees

- The algorithm works by first splitting the training set into two subsets using a single feature *k and a threshold* t_k

- How does it choose *k and* t_k?

- It searches for the pair (k, t_k) that produces the purest subsets (weighted by their size).

The CART Training Algorithm

- *CART cost function for classification:*

$$J(k, t_k) = \frac{m_{left}}{m} G_{left} + \frac{m_{right}}{m} G_{right}$$

- $G_{left/right}$ measures the impurity of left and right subset
- $m_{left/right}$ is the number of instances of left and right subset

The CART Training Algorithm

- Once the CART algorithm has successfully split the training set in two, it splits the subsets using the same logic, then the sub-subsets, and so on, recursively

- It stops recursing
  - once it reaches the maximum depth
  - or if it cannot find a split that will reduce impurity

CART Example

- Example problem: Predict if a person has heart disease or not based on three features using a Decision Tree

Chest Pain	Good Blood Circulation	Blocked Arteries	Heart Disease
NO	NO	NO	NO
YES	YES	YES	YES
YES	YES	NO	NO
	...		

Attribute #1 : Chest Pain

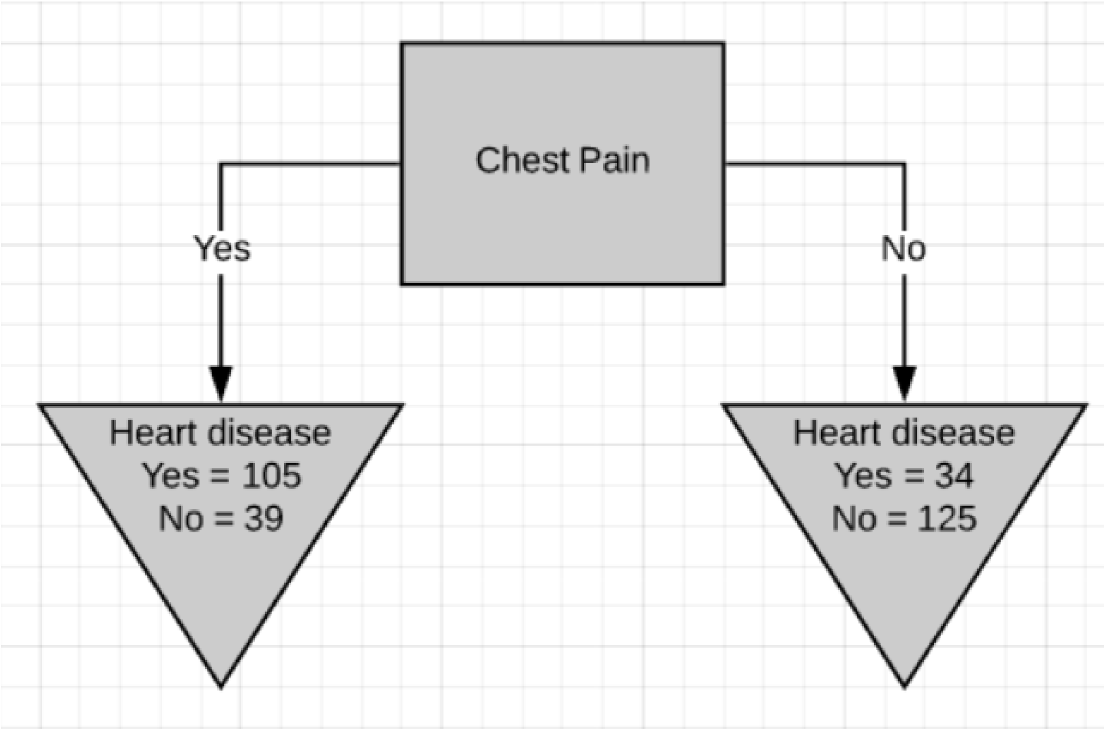


Fig.6-Chest pain as the root node

$$G_i = 1 - \sum_{k=1}^{n} {P_{i,k}}^2$$

$$= 1 - (\Pr(Yes))^2 - (\Pr(No))^2$$

- Left leaf Gini impurity

$$= 1 - (\frac{105}{105+39})^2 - \left(\frac{39}{105+39}\right)^2 = 0.395$$

- Right leaf Gini impurity

$$= 1 - (\frac{34}{125+34})^2 - \left(\frac{125}{125+34}\right)^2 = 0.336$$

Attribute #1 : Chest Pain

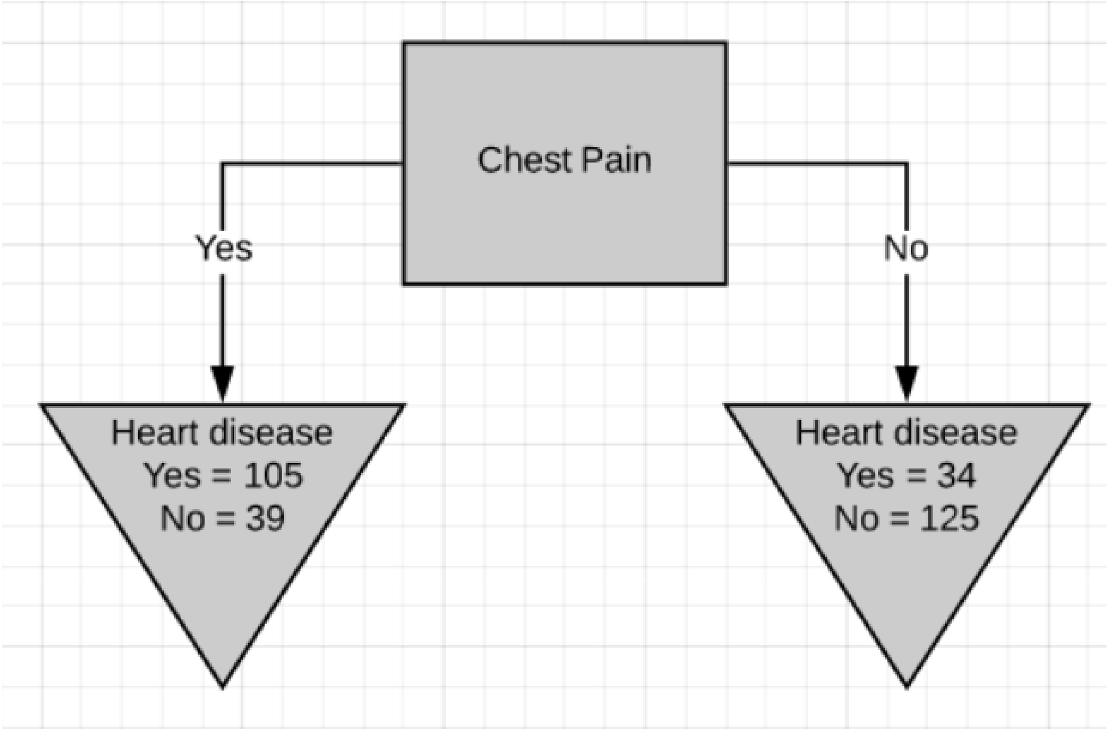
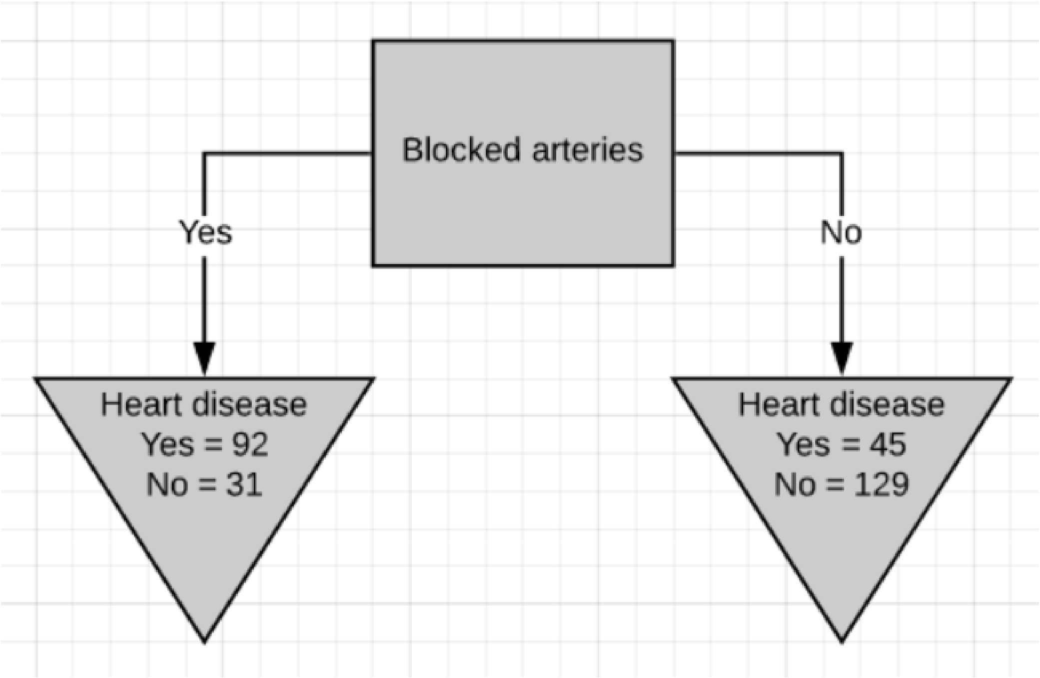


Fig.6-Chest pain as the root node

- Total Gini impurity – weighted average

$$= \left(\frac{144}{144+159}\right) * 0.395 + \left(\frac{159}{144+159}\right) * 0.336 = 0.364$$

Attribute #2 : Blocked Arteries



- Left leaf Gini impurity

$$= 1 - (\frac{92}{92+31})^2 - \left(\frac{31}{92+31}\right)^2 = 0.377$$

- Right leaf Gini impurity

$$= 1 - (\frac{45}{45+129})^2 - \left(\frac{129}{45+129}\right)^2 = 0.383$$

- Total Gini impurity – weighted average

$$= \left(\frac{123}{123+174}\right) * 0.377 + \left(\frac{174}{123+174}\right) * 0.383 = 0.381$$

Attribute #3: Good Blood Circulation

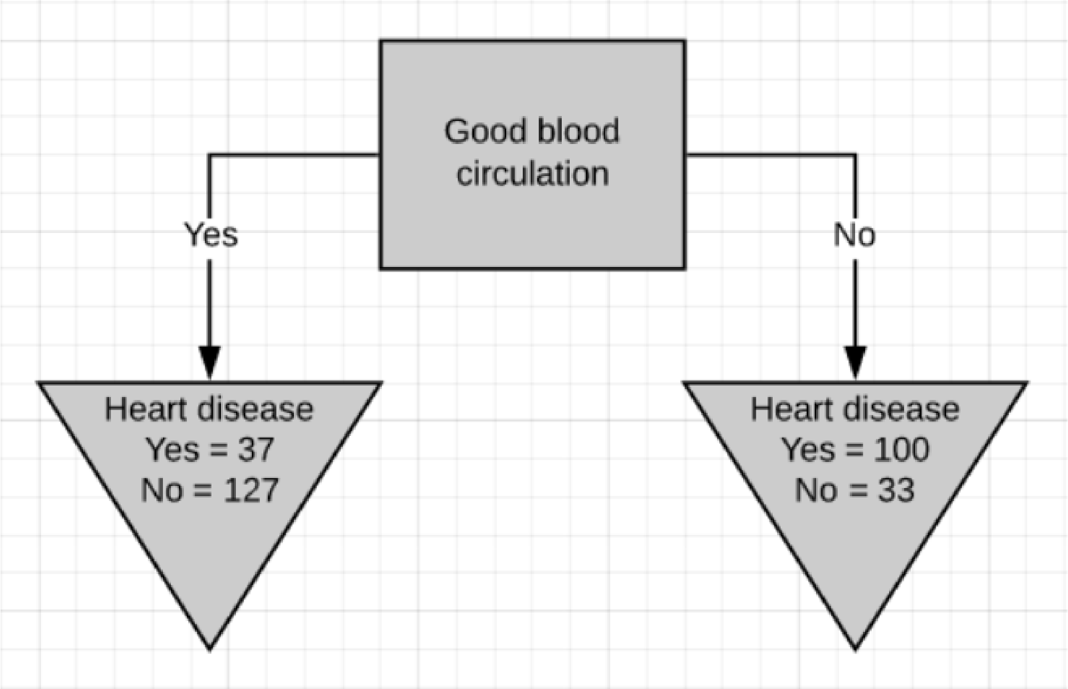


Fig.7-Good blood circulation as the root node

- Left leaf Gini impurity

$$= 1 - \left(\frac{37}{37 + 127}\right)^2 - \left(\frac{127}{37 + 127}\right)^2 = 0.349$$

- Right leaf Gini impurity

$$= 1 - \left(\frac{100}{100 + 33}\right)^2 - \left(\frac{33}{100 + 33}\right)^2 = 0.373$$

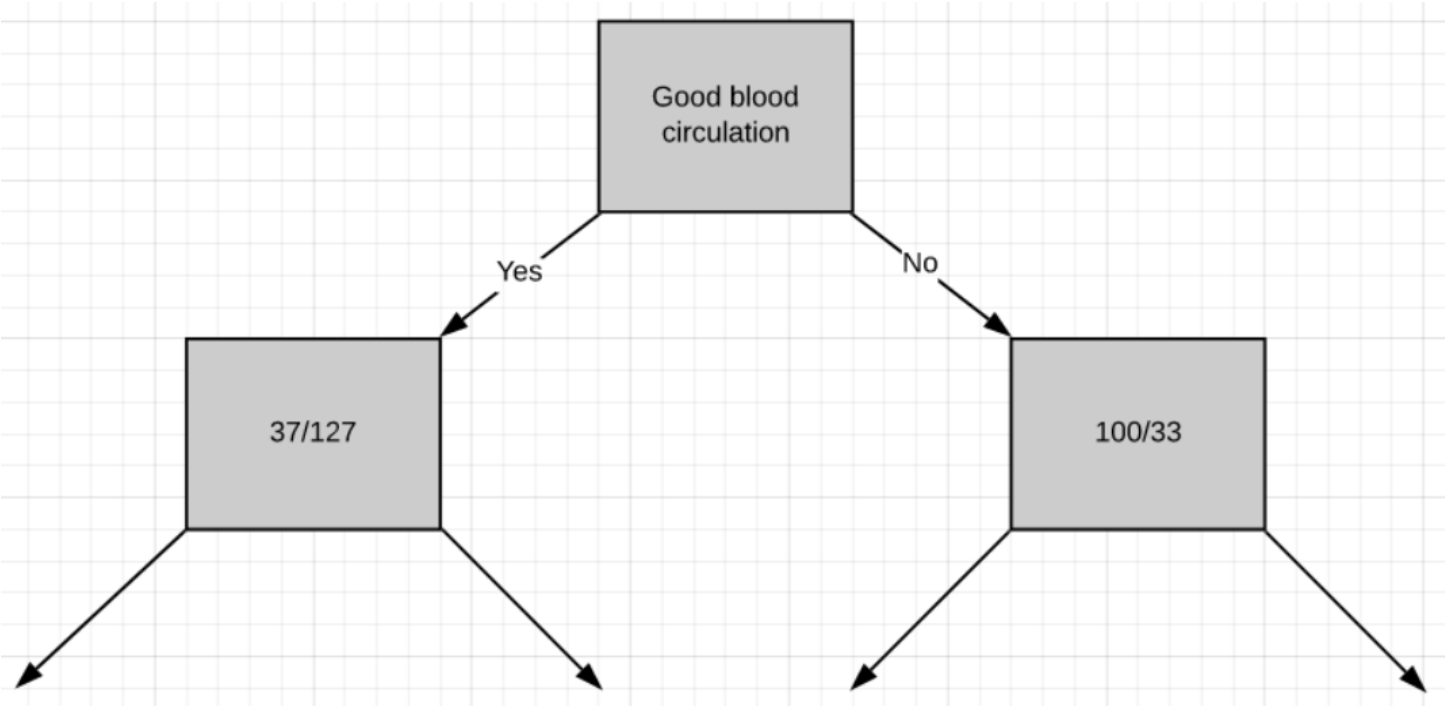
- Total Gini impurity – weighted average

$$= \left(\frac{164}{164 + 133}\right) * 0.349 + \left(\frac{133}{164 + 133}\right) * 0.373 = 0.360$$

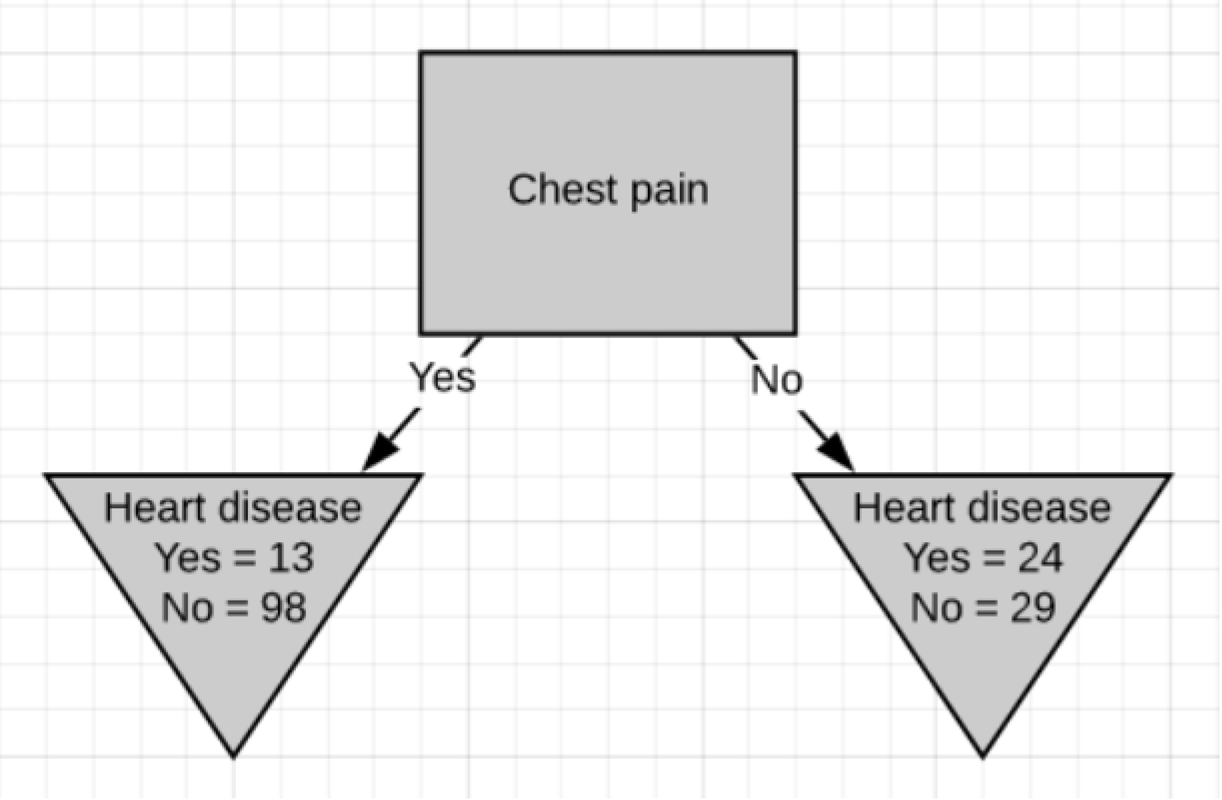
CART example

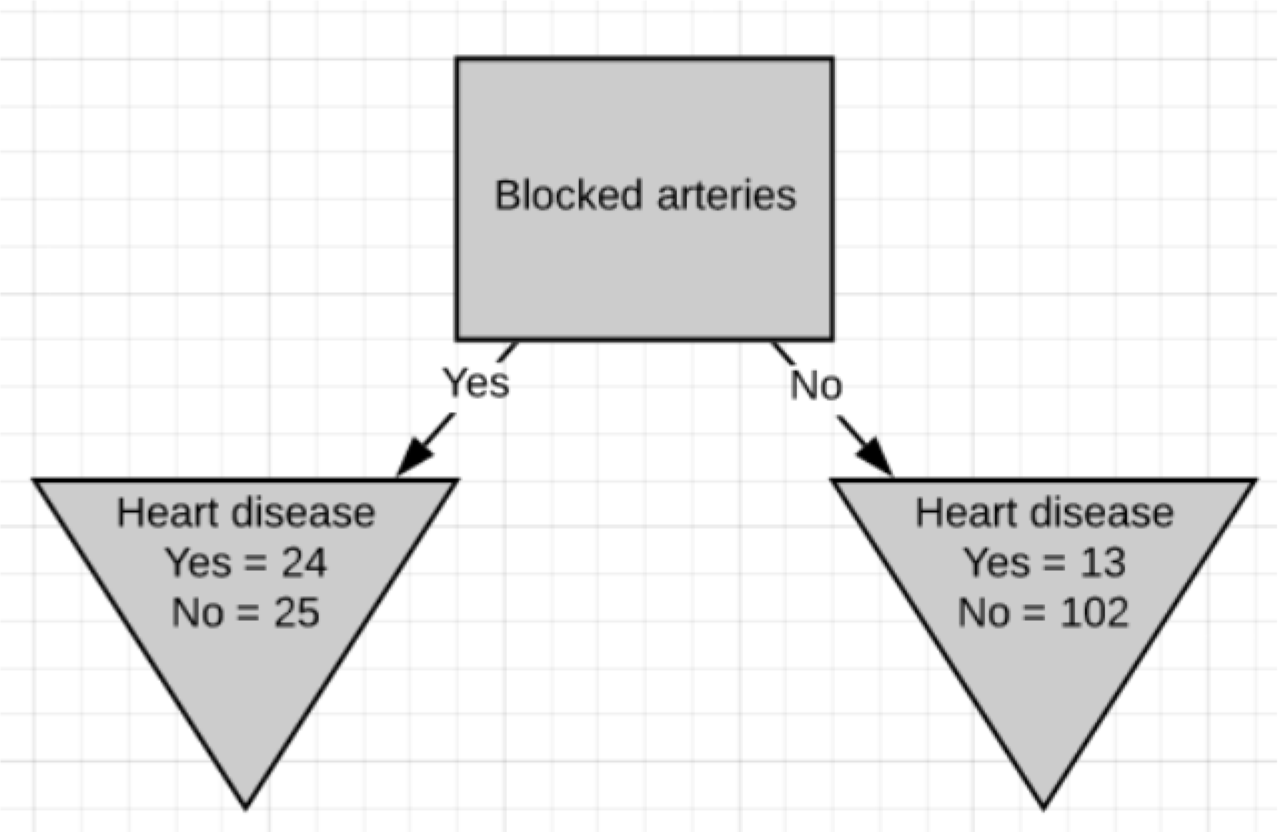
- Selecting GBC as root node divides the tree as follows

- Follow the same steps for the remaining internal nodes to find the decision tree

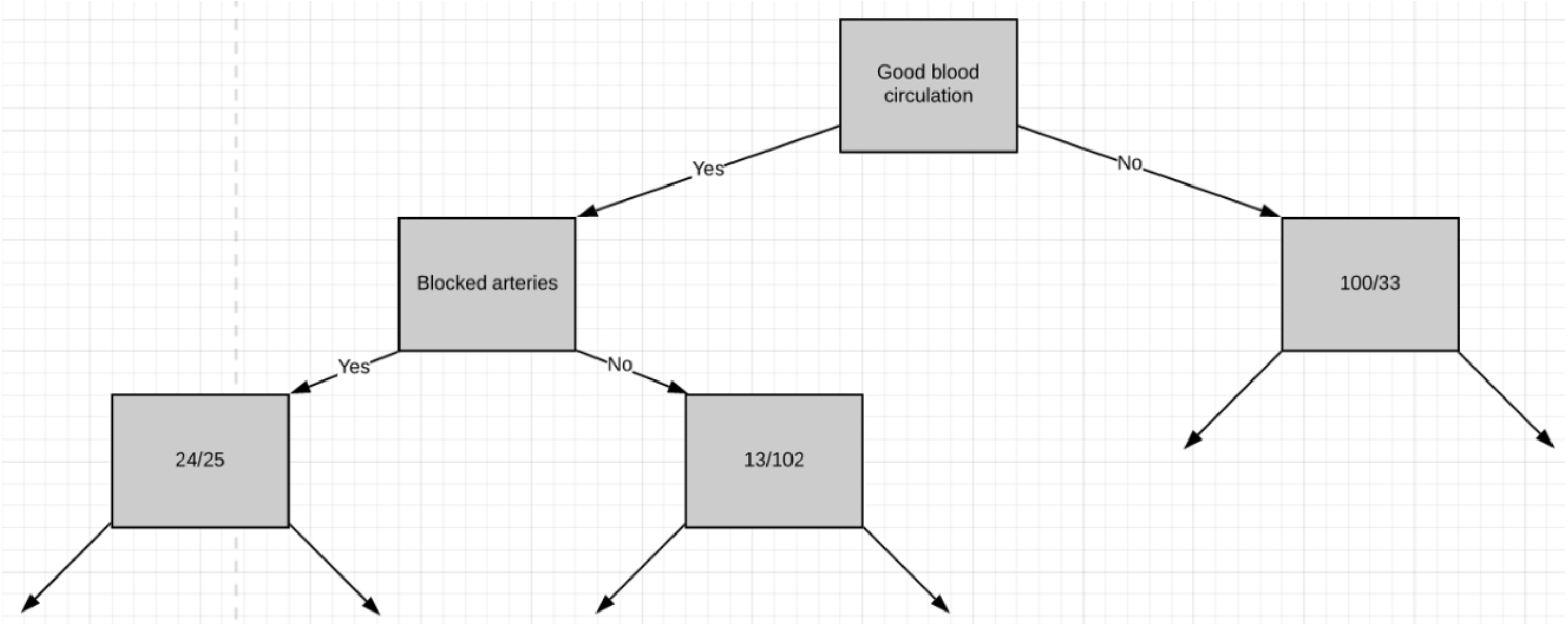


CART example

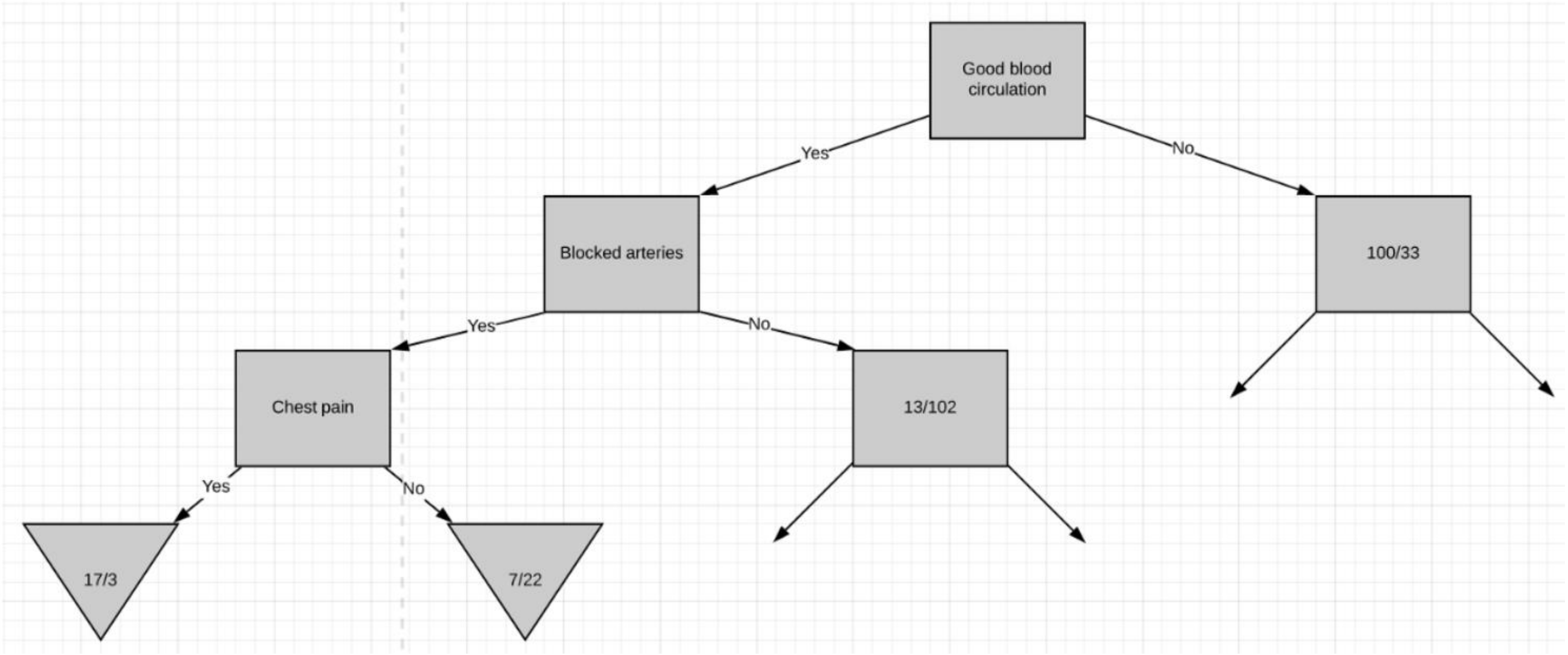




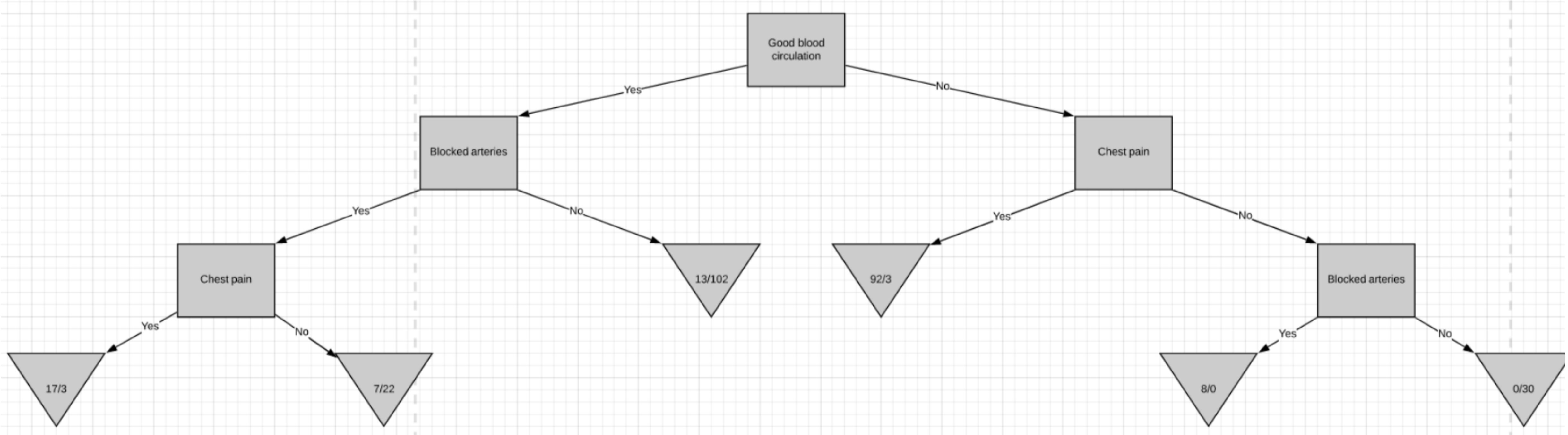
CART example



CART example



CART example



Decision Tree Hyperparameters sklearn

- *sklearn.tree.DecisionTreeClassifier*

class sklearn.tree.**DecisionTreeClassifier**(**, criterion='gini', splitter='best', max_depth=None, min_samples_split=2, min_samples_leaf=1, min_weight_fraction_leaf=0.0, max_features=None, random_state=None, max_leaf_nodes=None, min_impurity_decrease=0.0, class_weight=None, ccp_alpha=0.0*)

- max_depthint, default=None
- The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure
- min_samples_splitint or float, default=2
- The minimum number of samples required to split an internal node
- min_samples_leafint or float, default=1
- The minimum number of samples required to be at a leaf node
- max_featuresint, float or {“auto”, “sqrt”, “log2”}, default=None
- The number of features to consider when looking for the best split

Measure of Impurity: Entropy

- Entropy at a given node t,

$$Entropy = -\sum_{i=0}^{c-1} p_i(t) log_2 p_i(t)$$

- Here pi(t) is the frequency of class i at node t, and c is the total number of class.
- Maximum value can be log2c
  - When samples are equally distributed among classes
- Minimum value is 0
  - When all the samples belong to one class, implying most beneficial situation for classification
- Entropy based computations are quite similar to the Gini index computation